FAIR RESEARCH DATA

From principles to practical workflows



A workshop for making scientific data easier to discover, interpret, and reuse—responsibly.

THE Four FAIR PRINCIPLES



Findable | Data and metadata are easy to discover for humans and computers.



Accessible | Data can be retrieved using a clear, standardized access process (open or controlled).



Interoperable | Data can be integrated and used with other data and tools through shared standards and vocabularies.



Reusable | Data are well-described, with clear provenance, use conditions, and sufficient context for future use.

Why FAIR? Shared guardrails, less friction

Think of FAIR as a common language for describing and navigating data—not a mandate to remove every fence.





FAIR is about managing and standardizing the **guardrails** around data use and reuse to ensure a coherent, universal language for data communication, reduce burden, and support the entire **research data life cycle.**

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FAIR across the research data lifecycle

R -- Reusable
Provide clear use, conditions, provenance, and documentation

A -- Accessible
Define and implement appropriate access (e.g., open or controlled)

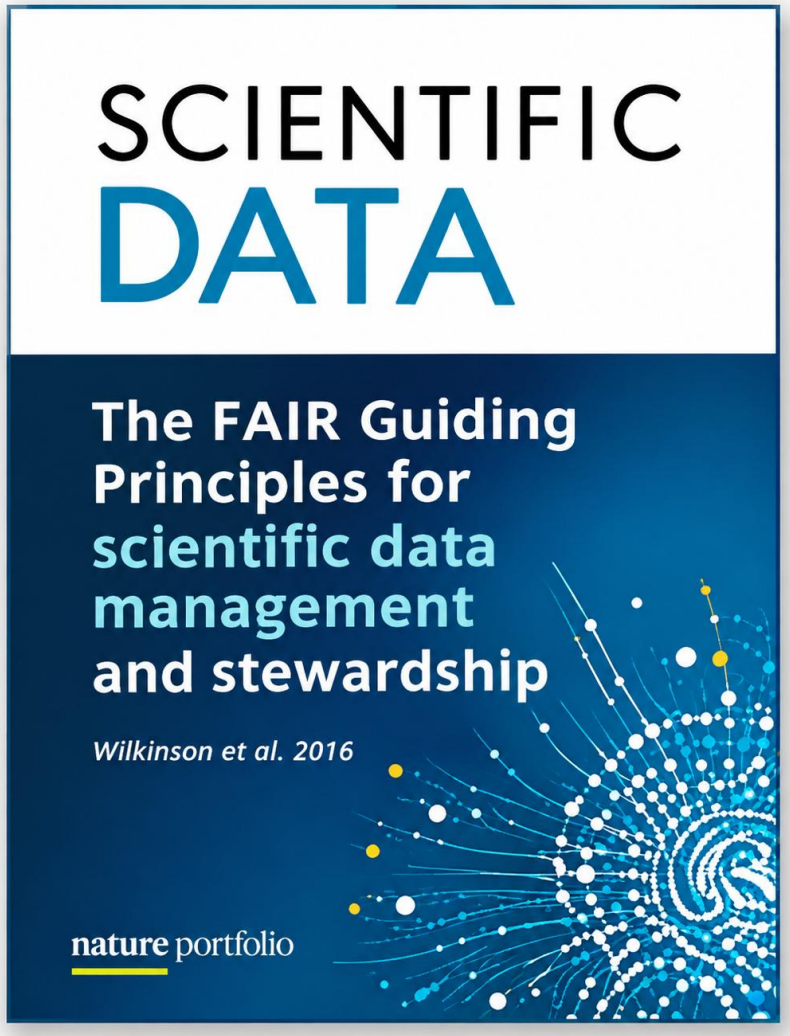
F -- Findable
Assign persistent identifiers, describe w. rich metadata, and register in a registry



I -- Interoperable
Use community standards, shared vocabularies, and compatible formats

All FAIR
Make each step more efficient, reproducible, and easier to build upon for future research.

The FAIR Guiding Principles (Wilkinson et al, 2016)



Original contribution

Wilkinson et al., Scientific Data (2016): 15 guiding principles for data and metadata stewardship.

Metadata matters

Descriptions and identifiers are essential even when the underlying data cannot be released.

Machine-actionability

The goal is for computational systems—not only people—to find and work with research resources.

Not a single technology

FAIR is a set of outcomes; repositories, schemas, identifiers and protocols are implementation choices.

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FAIR ≠ Open



OPEN DATA

- ✓ Freely available to anyone
- ✓ No access restrictions
- ✓ Often used for non-sensitive data
- ✓ Accessed w/o defined process
- ✓ May lack metadata

About “who” can access



FAIR CONTROLLED DATA

- ✓ Discoverable: Metadata is publicly available
- ✓ Accessible: Data has defined access process
- ✓ Interoperable: Uses community standards
- ✓ Reusable: Clear conditions for reuse
- ✓ Can be open or controlled-access(e.g., sensitive human data)

About “how” and “effect” access and use

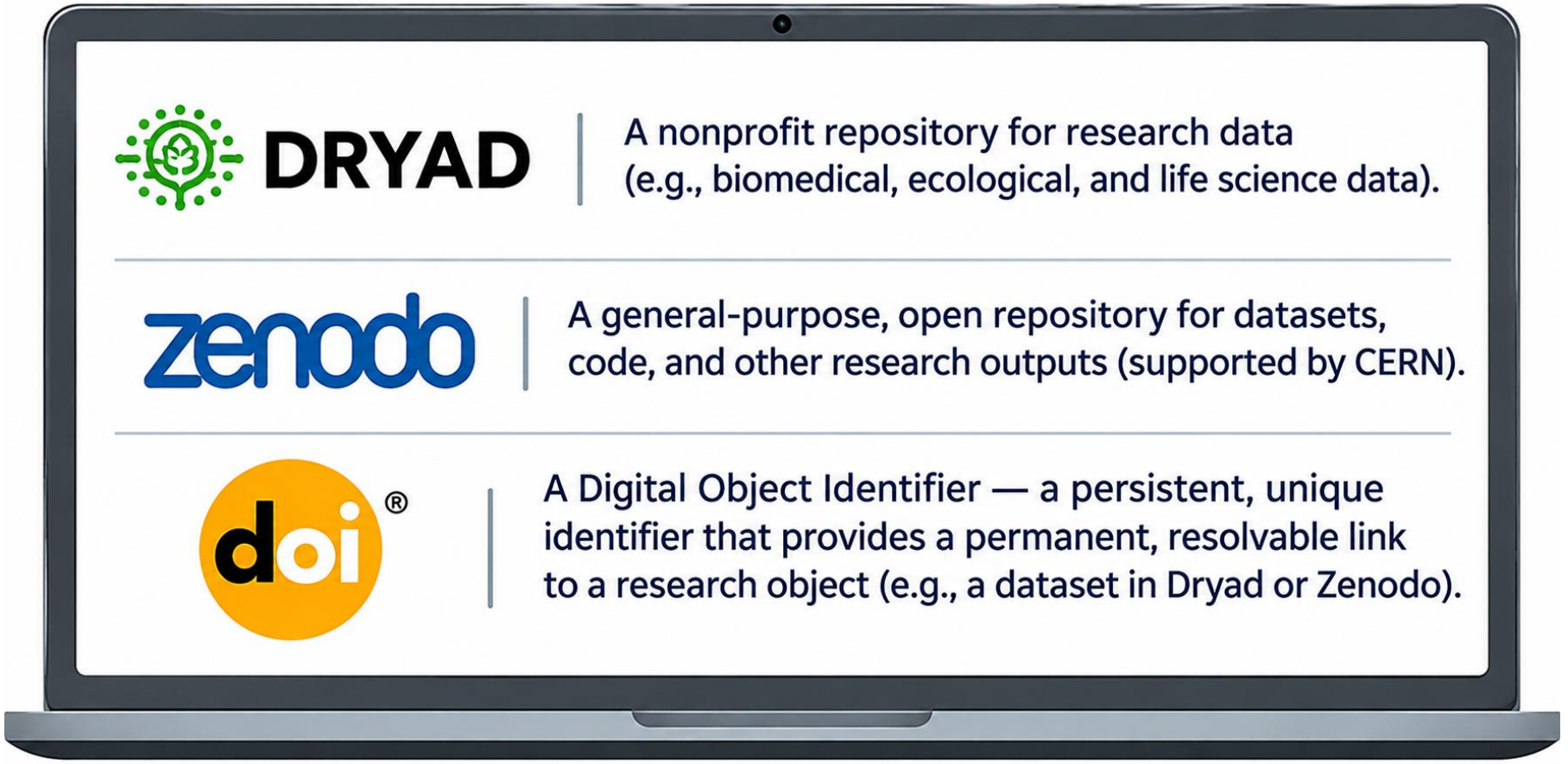
F — Findable: Make data discoverable and uniquely identifiable for humans and computers

F1	**Persistent, globally unique identifier**	A DOI or accession that distinguishes the dataset. Globally
F2	**Rich descriptive metadata**	Title, creator, methods, variables, dates, organism, context.
F3	**Metadata explicitly cites the data ID**	The catalog record clearly states which dataset it describes.
F4	**Indexed in a searchable resource**	A repository or catalog enables discovery by humans and software.

Examples: GEO / SRA / ENA accessions • PubMed PMIDs for articles • DOIs for deposited datasets

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DOIs, Dryad, and Zenodo: where findability becomes real



Trusted repositories and persistent identifiers that make research outputs findable, citable, and long-lived. Before deposit: check disciplinary repository expectations, privacy constraints, license, documentation, and versioning.

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Findable in practice: from folder to indexed record

01

Organize

README + data dictionary

02

Describe

Rich machine-readable metadata

03

Identify

Accession or DOI

04

Register

Repository + search index

Discovery path: search query → indexed metadata → persistent landing page → data or access instructions

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A — Accessible: Make data retrievable using clear, durable, retrievable rules

A1 Use a standardized, open, and universally implementable protocols (e.g., HTTP, HTTPS, FTP)

A2 Provide authentication and authorization process where necessary

A3 Ensure metadata remains accessible even if the data are access-restricted

A4 Clearly describe access conditions and any data use restrictions

Examples

Open access
(e.g., HTTP, HTTPS, FTP)

Controlled access
(e.g., dbGaP, EGA)

Access information
(e.g., data use agreement, request process)

Example: public cohort metadata → access request → approved secure workspace

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I — Interoperable:

Data can be combined, exchanged, and interpreted across tools and systems. Data can speak together.

I1 Use formal, shared and broadly applicable representation (e.g. standard file formats)

I2 Use shared vocabularies and ontologies (e.g., disease, phenotype, and experimental design)

I3 Include qualified references to other data (linked datasets, publications, identifiers, knowledge graphs)

I4 Use community standards that enable integration and computational use

Examples



Standard formats
(e.g., CSV, TSV, FASTQ, BAM)



Controlled vocabularies
(e.g., HPO, MONDO, GO)



Ontologies
(e.g., OBO Foundry)

Example: “Disease = ALL” → map to a defined ontology term + version + meaning → comparable across studies

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R — Reusable: Make the next study possible; ensure data is well-described and contextualized so they can be confidently and appropriately reused

I1 Provide rich, accurate, and relevant metadata (e.g., methods, context, and data provenance).

I2 Include a clear data-use/reuse license or conditions. Specify permitted use, restrictions, and citation expectations.

I3 Specify data provenance and processing steps (data generation, and analysis workflow documentation).

I4 Meet domain-relevant community standards and conventions (e.g. MIAME for expression data, MINSEQE for sequencing).

Examples



Documentation
(e.g., README, methods)



Licenses
(e.g., CC BY 4.0)



Provenance
(e.g., workflow, versioning)



Community standards
(e.g., MIAME, MINSEQE)

Reusable ≠ merely downloadable: a file without context may be impossible to interpret safely.

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A practical FAIR workflow for a research team



Build FAIRness into everyday research operations—not only at publication.

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Biomedical example: a pediatric genomics cohort

F — Findable

Public study record, accession, sample identifiers, and searchable metadata.

A — Accessible

Public description; individual-level files released only through approved access.

I — Interoperable

Common phenotype terms, assay schemas, reference genome version.

R — Reusable

Methods, QC, provenance, consent-compatible reuse conditions.

Sensitive does not mean un-FAIR. Good metadata and governance can coexist.

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Workshop exercise: FAIR audit in five minutes

- [ ] **Can another researcher FIND this dataset by a stable ID?**
- [ ] Can they understand HOW to request or retrieve it?
- [ ] Can software INTERPRET its fields and relationships?
- [ ] Can a new team REUSE it with confidence and permission?

Pick one dataset. Identify one improvement you could make this month.

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Takeaways & selected references

FAIR is a design practice: identify, describe, connect, govern, and preserve.

- Wilkinson et al. (2016). Scientific Data 3, 160018. nature.com/articles/sdata201618
- GO FAIR Principles. go-fair.org/fair-principles/
- GO FAIR Foundation. gofair.foundation/fair-principles
- DataCite DOI infrastructure. datacite.org | DOI Foundation. doi.org
- Dryad. datadryad.org | Zenodo. zenodo.org
- NLM FAIR / common data elements resources. nlm.nih.gov

Discussion: What is the smallest FAIR improvement with the largest impact on your next study?

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